

Program: Diploma in Automation and Robotics	
Course Code:	Course Title: Analog and Digital Electronics Lab
Semester: 3	Credits: 1.5
Course Category: Engineering Science	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To acquire skills in design and testing various BJT and FET based Circuits.
- To experiment on various Digital circuits using trainer kit and ICs.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Analog and Digital Electronics (concurrent with the semester)		Analog and Digital Electronics	3

Course Outcomes:

On completion of the course student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Experiment with various BJT based Circuits	9	Applying
CO2	Experiment with various FET and UJT based Circuits.	6	Applying
CO3	Identify different types of components and Digital ICs used for designing Combinational Circuits.	12	Applying
CO4	Experiment with different Sequential Circuits and Design Counter, Sequence Generators and Shift Registers.	12	Applying
	Lab Test	6	

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Experiment with various BJT circuits		
M1.01	Assemble and test a Common emitter amplifier circuit.	3	Applying
M1.02	Assemble and plot Common collector amplifier characteristics.	3	Applying
M1.03	Assemble and plot Common base amplifier parameters.	3	Applying
CO2	Experiment with various FET and UJT based circuits		
M2.01	Assemble and find the parameters of FET using common source amplifier.	3	Applying
M2.02	Design and setup UJT relaxation oscillator	3	Applying
	Lab Test-1	3	
CO3	Identify different types of components and Digital ICs used for designing combinational circuits.		
M3.01	Familiarize with digital trainer kit, IC tester and various digital ICs	3	Applying

M3.02	Demonstrate various types of adder, subtractor using basic gates and universal gates	3	Applying
M3.03	Using IC 7483 design a 4bit BCD adder	3	Applying
M3.04	Familiarize with MUX and DeMUX ICs	3	Applying
CO4	Experiment with different sequential circuits and design counter sequence generator and shift registers.		
M4.01	Familiarize with various flipflop ICs	3	Applying
M4.02	Design and implement various shift registers using D flipflop	3	Applying
M4.03	Design and implement 4 bit up asynchronous up and down counter using flipflops	3	Applying
M4.04	Design and implement synchronous sequence generators using flipflops.	3	Applying
	Lab Test– II	3	

Text/Reference

T/R	Book Title/Author
T1	B. L. Thereja and A. K. Thereja, “Textbook of Electrical Technology: Part 1 - Basic Electrical Engineering in S.I. Units”, S. Chand Publication, 2012
T2	N.N. Bhargava, D.C. Kulshreshtha S.C. Gupta, “Basic Electronics and Linear Circuits”, Tata McGraw Hill Education, 1983
R3	A. Anand Kumar, “Fundamentals of Digital Circuits”, Prentice Hall India Pvt.Ltd., 2014

Online Resources

Sl No	Website Link
1	www.electrical4u.com
2	https://www.digimat.in/nptel/courses/video/108101091/L01.html
3	https://www.digimat.in/nptel/courses/video/108105112/L01.html